

Solar Powered Cooling System

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Project Overview

- Portable solar-powered system for personal cooling in outdoor environments.
- Utilizes renewable energy as primary source.
- Designed for portability, energy efficiency, and minimal user interaction.
- Supports continued operation under lowlight conditions through battery storage.
- Demonstrates integration of a small-scale renewable energy system with embedded control in a compact application.

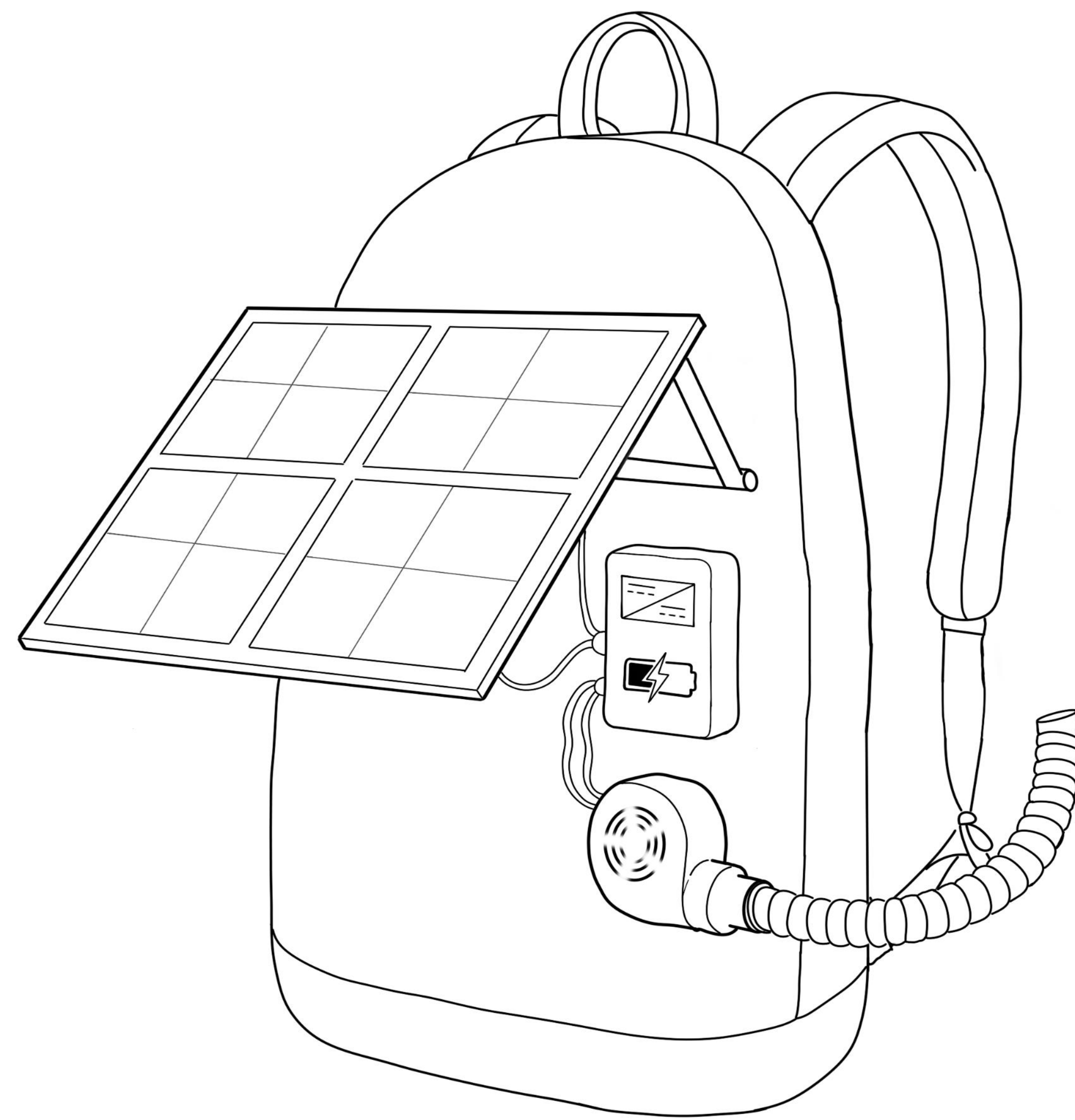


Figure 1: Portable system concept art

Motivation

- Reduce dependence on grid charging and disposable batteries.
- Promote accessible renewable energy in practical applications.

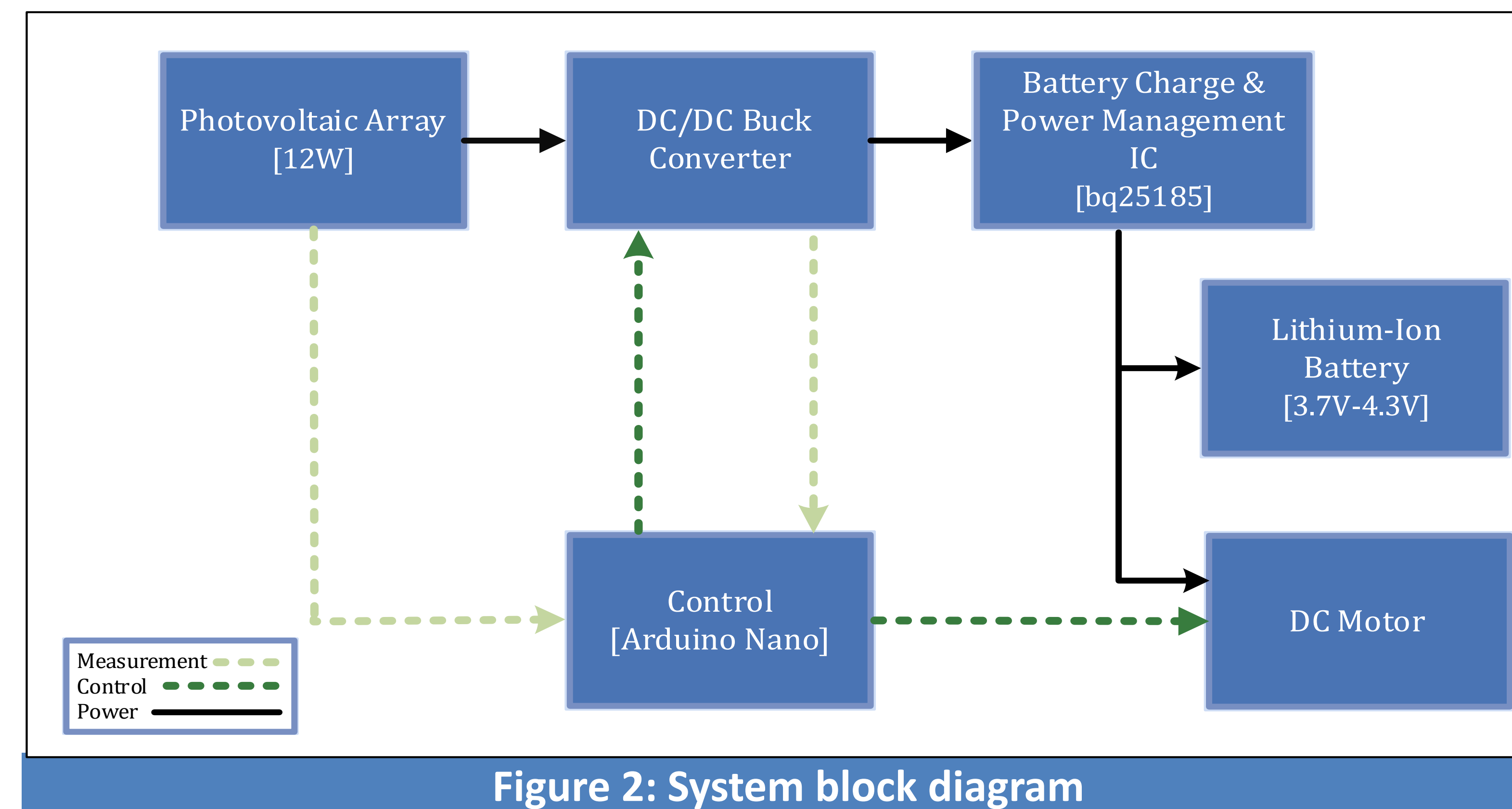


Figure 2: System block diagram

System Design

- Solar array provides input energy source.
- DC/DC buck converter regulates power transfer from solar source.
- BQ25185 IC manages lithium-ion battery charging and system power distribution between battery and fan
- Arduino Nano coupled with INA226 sensors perform sensing, PWM generation and converter control.
- Integrated control enables autonomous operation under varying sunlight and load conditions.
- DC motor provides airflow to user.

Results

- Buck converter successfully designed and implemented in hardware.
- Stable output voltage regulation to $6.0 \pm 0.1V$ (Figure 3).
- System demonstrated reliable operation under battery and fan loading conditions.
- Stable converter startup achieved using a minimum load resistor of 22Ω
- Arduino Nano and INA226 successfully monitors output voltage and current.
- Successful operation of BQ25185 managed battery charging and power delegation to fan.
- Custom 3D-printed fan housing for directional cooling.
- Portable 3D-printed enclosure for protection.

Challenges and Solutions

During hardware testing, the buck converter was found to require a minimum load to achieve reliable startup and maintain output voltage regulation while supplying the battery management IC. Experimental testing with a DC electronic load was used to determine the resistance value and power consumption. A 22Ω resistor was implemented and found to consume 1.64W. Since the resistor is only required upon start up, a low-side MOSFET switch was added to disconnect the resistor once the converter output reached the regulated 6V operating point.

Additional testing identified excessive voltage transients caused by the MOSFET switching. Increasing the gate resistance between the IR2110 gate driver and converter MOSFET reduced switching-induced voltage spikes seen on the output. As shown in Figure 3, the resulting output voltage remains close to the desired voltage.

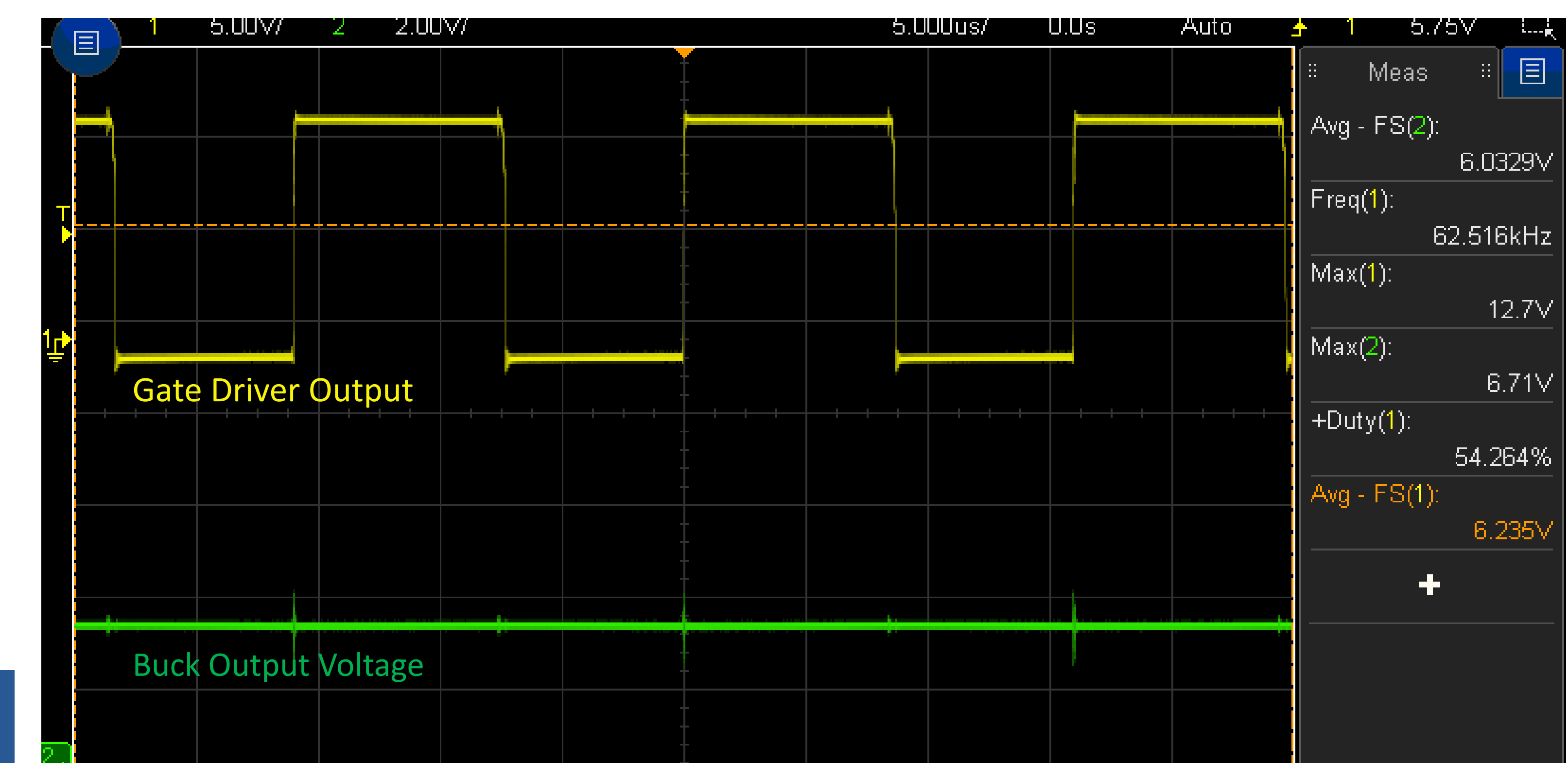


Figure 3: Gate driver output and converter output voltage waveforms

Future Direction

- Implement system on a custom PCB to improve portability and performance.
- Implement MPPT for maximum solar power extraction.
- Perform thermal analysis of converter and battery during extended operation.

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